



UCSC

University of Colombo, Sri Lanka

University of Colombo School of Computing

BIT

**DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**

Academic Year 2022— 3rd Year Examination — Semester 6

EN6106 — Emerging Topics in Information Technology

Multiple Choice Question Paper

(1 Hours)

Important Instructions

- The duration of the paper is **1 Hours**.
- The medium of instructions and questions is English.
- This paper has **25 questions** on **8 pages**. Answer **all** questions.
- All questions are of the **MCQ** (Multiple Choice Questions) type.
- Each question will have **5 (five)** choices with **one or more** correct answers.
- This paper consists of 100 marks and all the questions will carry equal marks.
- There will be a penalty for incorrect responses to discourage guessing.
- The mark given for a question will vary from -1 (All the incorrect choices are marked & no correct choices are marked) to +1 (All the correct choices are marked & no incorrect choices are marked). However, **the minimum mark per question would be zero**.
- Answers should be marked on the **special answer sheet** provided.
- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor/invigilator immediately.
- Mark the correct choices on the question paper first and then transfer them to the given answer sheet which will be machine marked. **Please completely read and follow the instructions given on the other side of the answer sheet before you shade your correct choices.**
- Calculators are **not** allowed.
- *All Rights Reserved*. This question paper can NOT be used without proper permission from the University of Colombo School of Computing.

- 1) A monolithic architecture is the traditional model for software program design. What statement(s) best describes the *monolithic architecture* of a software system?

- (a) All functionalities are implemented and deployed into a single software application.
- (b) All functionalities are implemented separately but deployed into a single software application.
- (c) All functionalities are implemented into a single application but deployed as separate software applications.
- (d) Monolithic software application has many fine-grained and independent services
- (e) Monolithic software applications use RESTful web services but are not supported for SOAP web services

- 2) What is/are benefit(s) of having a *monolithic software system*?

- (a) Practice agile development and delivery methodologies
- (b) Easy upgrading and adding new features
- (c) Easy to scale
- (d) Simplicity of having one code base
- (e) Speedy development and deployment

- 3) Microservices refers to an architectural style for developing applications. What is/are the feature(s) of a *microservices* application?

- (a) Single application as a suite of fine-grained and independent services
- (b) Services can be developed and deployed independently
- (c) Cannot be hosted on web servers
- (d) Can be implemented using Spring Boot framework
- (e) RESTful API can be used to connect frontend and backend microservices

- 4) One significant advantage of microservices is decentralized data management. What is/are the feature(s) of *Decentralized Data management* in Microservices?

- (a) Each microservice can have a private database to persist the data that requires to implement the business functionality offered by it.
- (b) In some business scenarios, you might have to update several databases for a single transaction.
- (c) In a microservice application, it is impossible to configure a single database where tables are separated by clearly defined, logical boundaries and owned by specific services.
- (d) Decentralized database management is the system in which all the data is stored and managed in a single unit.
- (e) A given microservice can only access the dedicated private database but not the databases of other microservices.

- 5) Containers and virtual machines make your applications independent from your IT infrastructure resources. What is/are the correct statement(s) about *Virtual Machines (VMs)* and *Docker containers*?

- (a) Each VM has an OS inside each VM, and it is called a host OS. Host OS can be any OS, like Linux or Windows.
- (b) Docker containers host on a single physical server with a host OS, which shares among them. Sharing the host OS between containers makes them light and increases the boot time.
- (c) Docker containers are suitable for running multiple applications over a single OS kernel. Virtual machines are needed if the applications or services run on different OS.
- (d) Docker lets you separate your applications from your infrastructure to deliver software quickly.
- (e) Virtual machines allow developers to build, test, and deploy code within simulated environments

- 6) *Data scientist* and *data engineer* are the two main professions in bigdata projects. What is/are the correct statements about data scientists and data engineers?

- (a) Data scientist is operation focused while data engineer is business focused
- (b) Data scientist analyses data and builds ML models while the data engineer provides data required for ML models
- (c) Data scientists clean and prepare data while data engineers move data
- (d) Data scientists require knowledge of mathematics, statistics, and computer science, while data engineers need knowledge of big data technologies and programming
- (e) Data scientists create Extract - Transform - Load pipeline while data engineers produce insights and recommendations for improving the business

- 7) Python is a programming language widely used by Data Scientists. What are the *libraries* in Python that can be used in data analysis?

- (a) Pandas
- (b) Rqueue
- (c) Numpy
- (d) SciFi
- (e) Simple Flat Mapper

- 8) *Data visualization* is the representation of data through the use of common graphics, such as charts, plots, infographics, and animations. What is/are the benefit(s) of data visualization?

- (a) Identify outliers
- (b) Easily understand casual relationships
- (c) Visualize patterns
- (d) Easy way to share information for non-technical audiences
- (e) Data visualization tools show insights that may be missed in traditional reports

- 9) Several steps can be taken to use data science projects to *extract meaning from data*. Identify the important steps of this process.

- (a) Data Cleaning
- (b) Data Interpretation
- (c) Data Modeling
- (d) Data Serialization
- (e) Data Plumbing

- 10) *Data pipelining tools* and solutions come in many forms, such as batch, cloud, and real-time. What is/are the common requirement(s) of these tools?

- (a) Extract data from multiple relevant data sources
- (b) Byzantine fault tolerance (BFT)
- (c) Clean, alter, and enrich the data so it can be ready for analysis
- (d) Load the data to a single source of information, usually a data lake or a data warehouse
- (e) Exploratory data analysis

- 11) AI is focused on creating systems/machines with human-like thinking and reasoning abilities and bridging the gap between human intelligence and computational systems. What is/are examples of *AI paradigms* proposed to model AI systems/machines?

- (a) Ontological
- (b) Epistemological
- (c) Bayesian
- (d) Axiological
- (e) Evolutionary

- 12) What can be considered as an example(s) for an *expert system*?

- (a) SMTP Server
- (b) FTP Server
- (c) Version Control System
- (d) Cancer Decision Support System
- (e) Microsoft Project

- 13) Artificial Intelligence can be divided mainly into two types based on *capabilities*. They are *Strong AI* and *Weak AI*. Which of the following statement(s) is/are true?

- I. In Strong AI, machines can solve complex problems like intelligent humans.
- II. Weak AI, also known as narrow AI, focuses on performing a specific task
- III. There are no real examples of strong AI because it is a hypothetical theory.

- (a) Only I is correct
- (b) I and II are correct
- (c) Only II is correct
- (d) II and III are correct
- (e) I, II, and III are correct

14) Based on the *functionality*, AI can be classified into different types. Which of the following is/are *NOT* a type of AI?

- (a) Reactive Machines
- (b) Limited Memory
- (c) Short-Term Memory
- (d) Theory of Mind
- (e) Self-Awareness

15) To obtain conclusions from data using artificial intelligence (AI), two commonly utilized approaches and *Rules-based AI* and *Machine Learning based AI*. In relation to those two approaches, which of the following statement(s) can be considered true?

- I. Rules-based AI systems are suited for projects and applications that require small amounts of data and simple, straightforward rules.
- II. Machine Learning based AI systems are commonly used when large volumes of relevant data records are available for making more accurate predictions.
- III. There are some systems available that combine rules-based AI systems with ML systems

- (a) Only I is correct
- (b) I and II are correct
- (c) Only II is correct
- (d) II and III are correct
- (e) I, II, and III are correct

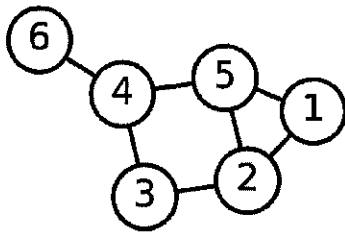
16) Social Networks can be represented mainly using *vertices* and *edges*. Which of the following statement(s) can be considered as false?

- (a) Two vertices are adjacent if they are connected to each other through an edge
- (b) An undirected edge is an ordered pair of nodes that can be represented graphically as an arrow drawn between the nodes.
- (c) A directed edge disregards any sense of direction and treats both nodes interchangeably
- (d) Out Degree is the number of edges leaving a vertex
- (e) In degree is the number of edges entering a vertex

17) What are well-known *centrality measures* in Social Network Analysis?

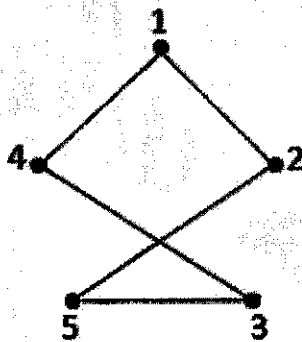
- (a) Degree
- (b) Betweenness
- (c) Closeness
- (d) Openness
- (e) Eigenvector

18) What kind of data structure shows by the following figure?



- (a) Undirected graph
- (b) Directed graph
- (c) Multigraph
- (d) Complete Graph
- (e) Singleton graph

19) Which *adjacency matrix* shows the correct representation of the following graph?



(a)

	1	2	3	4	5
1		1		1	
2	1				1
3				1	1
4	1		1		
5		1	1		

(b)

	1	2	3	4	5
1		1		1	
2	1				1
3		1			1
4	1		1		
5		1	1		

(c)

	1	2	3	4	5
1		1		1	
2				1	1
3		1			1
4	1		1		
5		1	1		

(d)

	1	2	3	4	5
1		1		1	
2	1				1
3		1			1
4	1		1		
5		1		1	

(e)

	1	2	3	4	5
1		1		1	
2	1				1
3		1			1
4	1				1
5		1	1		

20) *Data Management* in social network analysis is crucial for:

- (a) Protecting sensitive information, adhering to ethical guidelines and regulations.
- (b) Transparent documentation for sharing, collaboration, and replicability
- (c) Structuring data coherently to analyze relationships in the network effectively
- (d) Standardizing data, removing errors and duplicates, and ensuring trustworthy results
- (e) To deal with potential biases like selection bias

21) *Forensic science* is made up of many specialized areas, including

- (a) Forensic Psychology
- (b) Forensic Chemistry
- (c) Forensic Ontology
- (d) Criminalistics
- (e) Forensic Physics

22) Digital forensics is a branch of forensic science that focuses on identifying, acquiring, processing, analyzing, and reporting data stored electronically. What can be considered as the evidence gathered under Digital forensics?

- (a) DNA tests
- (b) Internet browser histories
- (c) Metadata
- (d) Ballistics
- (e) Contents of computer memory

23) With digital evidence, investigators can quickly identify suspects, track their behaviours/movements, and gather evidence to support a prosecution. What are the roles of Digital Forensics in *Law Enforcement*?

- (a) Digital Evidence Collection
- (b) Fraud Investigations
- (c) Cybercrime Investigations
- (d) Private Investigations
- (e) Terrorism Investigations

24) Digital forensic investigation is a process with set of phases. Identify the *phases* related to Digital forensic investigation from the list below.

- (a) Identification
- (b) Examination
- (c) Digitalization
- (d) Analysis
- (e) Archiving

25) What statement(s) is/are true related to the *Chain of Custody*?

- (a) A way to document the handling and movement of evidence in a digital forensic investigation
- (b) Looks for patterns, connections, and other meaningful relationships to develop an understanding of the case
- (c) Helps prevent tampering, contamination, or loss of evidence
- (d) Starts when the evidence is first collected and ends when presented in court
- (e) Use a variety of techniques to examine digital evidence, including specialized software tools and manual analysis
